

1. Write a shell script to generate mark- sheet of a student. Take 3 subjects, calculate and display total marks, percentage and class obtained by the student

The screenshot shows a terminal window with two parts. The top part is a nano editor editing a file named 'program.sh'. The script prompts for marks of three subjects, calculates the total and percentage, and assigns a class based on the percentage. The bottom part shows the terminal execution of the script, where the user enters marks of 45, 67, and 78, resulting in a total of 190, a percentage of 63%, and a class of 'First Class'.

```
GNU nano 8.7 program.sh
#!/bin/bash
#!/bin/bash

echo "Enter marks of Subject 1:"
read m1
echo "Enter marks of Subject 2:"
read m2
echo "Enter marks of Subject 3:"
read m3

total=$((m1 + m2 + m3))
percentage=$((total / 3))

echo "Total Marks = $total"
echo "Percentage = $percentage%"

if [ $percentage -ge 75 ]
then
    echo "Class: Distinction"
elif [ $percentage -ge 60 ]
then
    echo "Class: First Class"
elif [ $percentage -ge 50 ]
then
    echo "Class: Second Class"
else
    echo "Class: Fail"
fi
```

```
mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$ nano marksheet.sh
mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$ chmod +x marksheet.sh
mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$ ./marksheet.sh
Enter marks of Subject 1:
45
Enter marks of Subject 2:
67
Enter marks of Subject 3:
78
Total = 190
Percentage = 63%
Class: First Class
mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$
```

2. Write a menu driven shell script which will print the following menu and execute the given task.

- Display calendar of current month
- Display today's date and time
- Display usernames those are currently logged in the system
- Display your terminal number

```
MINGW64/c/Users/mankar/Desktop
GNU nano 8.7 menu.sh
#!/bin/bash

echo "-----"
echo "      MENU"
echo "-----"
echo "1. Display calendar of current month"
echo "2. Display today's date and time"
echo "3. Display usernames currently logged in"
echo "4. Display your terminal number"
echo "-----"
echo "Enter your choice:"
read ch

case $ch in
1) cal ;;
2) date ;;
3) who ;;
4) tty ;;
*) echo "Invalid Choice" ;;
esac

AG Help      AO Write Out  AF Where Is  AK Cut       AT Execute   AC Location  M-U Undo    M-A Set Mark M-] To Bracket M-B Previous + Back
AX Exit      AR Read File  \ Replace    AU Paste     AJ Justify   A/ Go To Line M-E Redo    M-6 Copy     AB Where Was  M-F Next    + Forward
```

```
mankar@LAPTOP-TUALCH80 MINGW64 ~
$ ./menu.sh

-----
      MENU
-----
1. Display calendar of current month
2. Display today's date and time
3. Display usernames currently logged in
4. Display your terminal number
-----
Enter your choice:
4
/dev/pty0

mankar@LAPTOP-TUALCH80 MINGW64 ~
$
```

3. Write a shell script which with generate first n Fibonacci numbers like: 1,1,2, 3, 5,13

```
MINGW64/c/Users/mankar/Desktop
GNU nano 8.7 fibonacci.sh
#!/bin/bash

echo "Enter the number of terms:"
read n

a=0
b=1

echo "Fibonacci Series:"
for ((i=1; i<=n; i++))
do
    echo -n "$a "
    c=$((a + b))
    a=$b
    b=$c
done
echo

AG Help      AO Write Out  AF Where Is  AK Cut       AT Execute   AC Location  M-U Undo    M-A Set Mark M-] To Bracket M-B Previous + Back
AX Exit      AR Read File  \ Replace    AU Paste     AJ Justify   A/ Go To Line M-E Redo    M-6 Copy     AB Where Was  M-F Next    + Forward
```

```

mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$ nano fibonacci.sh

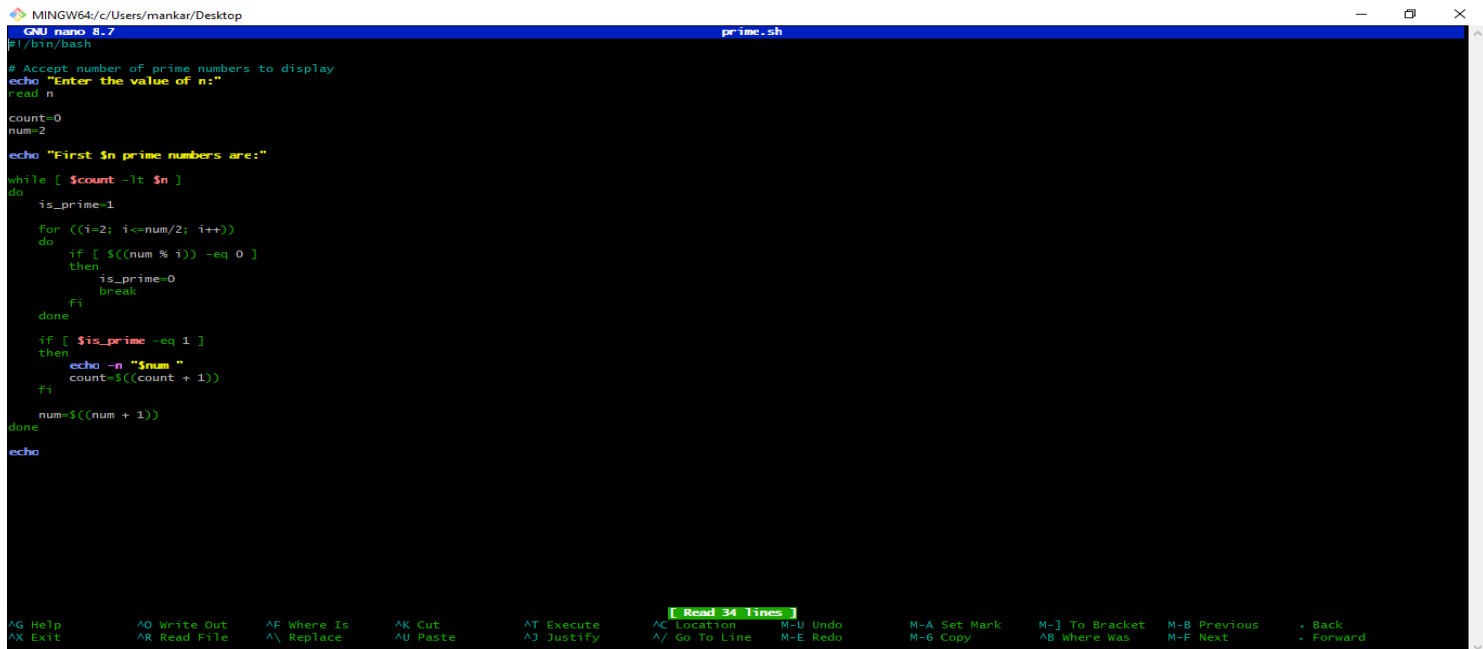
mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$ chmod +x fibonacci.sh

mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$ ./fibonacci.sh
Enter number of terms:
5
Fibonacci Series:
0 1 1 2 3

mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$

```

4. Write a shell script which will accept a number b and display first n prime numbers as output.



```

MINGW64/c/Users/mankar/Desktop
GNU nano 2.7.1 prime.sh
#!/bin/bash
# Accept number of prime numbers to display
echo "Enter the value of n:"
read n
count=0
num=2
echo "First $n prime numbers are:"
while [ $count -lt $n ]
do
    is_prime=1
    for ((i=2; i<=num/2; i++))
    do
        if [ $(($num % i)) -eq 0 ]
        then
            is_prime=0
            break
        fi
    done
    if [ $is_prime -eq 1 ]
    then
        echo -n "$num "
        count=$((count + 1))
    fi
    num=$((num + 1))
done
echo

```

Read 34 Lines

AG Help AO Write Out AF Where Is AK Cut AT Execute AC Location M-U Undo M-A Set Mark M-] To Bracket M-B Previous . Back
 AX Exit AR Read File A\ Replace AU Paste AJ Justify A/ Go To Line M-E Redo M-6 Copy M-] To Bracket M-B Previous . Back
 M-F Next . Forward

```

mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$ chmod +x prime.sh
bash: chmod: command not found

mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$ ./prime.sh
Enter the value of n:
2
First 2 prime numbers are:
2 3

mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$ |

```

5. Write menu driven program for file handling activity.

- Creation of file
- Write content in the file
- Upend file content
- Delete file content

```

MINGW64/c/Users/mankar/Desktop
GNU nano 8.7 filehandling.sh
#!/bin/bash

echo "
-----
FILE HANDLING MENU
-----
1. Create a File
2. Write content to file
3. Append content to file
4. Delete file content
-----
Enter your choice:"
read ch

echo "Enter file name:"
read fname

case $ch in
1)
touch $fname
echo "File created successfully"
;;
2)
echo "Enter content to write:"
read content
echo "$content" > $fname
echo "Content written to file"
;;
3)
echo "Enter content to append:"
read content
echo "$content" >> $fname
echo "Content appended to file"
;;
4)
> $fname
echo "File content deleted"
;;
*)
echo "Invalid choice"
;;
esac

AG Help      AD Write Out  AF Where Is  AK Cut       AT Execute   AC Location  M-U Undo    M-A Set Mark M-J To Bracket M-B Previous + Back
AK Exit      AR Read File  AL Replace   AU Paste     AJ Justify   A/ Go To Line M-E Redo    M-G Copy     AB Where Was  M-F Next    + Forward

```

```

mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$ chmod +x filehandling.sh

mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$ ./filehandling.sh
-----
FILE HANDLING MENU
-----
1. Create a file
2. Write content to file
3. Append content to file
4. Delete file content
-----
Enter your choice:
2
Enter file name:
laxmi
Enter content to write:
OS repository
Content written to file

mankar@LAPTOP-TUALCH80 MINGW64 ~/Desktop
$

```